CHAPTER 3

Classification of Elements and Periodicity in Properties

Modern Periodic Law, Nomenclature of Elements

1. The IUPAC name of an element with atomic number 119 is. *(2022)*

a. ununcotium b. ununennium
c. unnilennium d. unununnium

2. Identify the incorrect match *(2020)*

Name		IUPAC Official Name	
(A)	Unnilunium	(i)	Mendelevium
(B)	Unniltrium	(ii)	Lawrencium
(C)	Unnilhexium	(iii)	Seaborgium
(D)	Unununnium	(iv)	Darmstadtium

a. (B), (ii) b. (C), (iii)
c. (D), (iv) d. (A), (i)

Electronic Configurations and Types of Elements (s, p, d and f-Blocks)

3. The element Z = 114 has been discovered recently. It will belong to which of the following family group and electronic configuration? *(2017-Delhi)*

a. Nitrogen family, [Rn] $5f^{14}6d^{10}7s^27p^6$
b. Halogen family, [Rn] $5f^{14}6d^{10}7s^27p^5$
c. Carbon family, [Rn] $5f^{14}6d^{10}7s^27p^2$
d. Oxygen family, [Rn] $5f^{14}6d^{10}7s^27p^4$

4. The number of d-electrons in Fe^{2+} (Z = 26) is not equal to the number of electrons in which one of the following? *(2015)*

a. p-electrons in Cl (Z = 17)
b. d-electrons in Fe (Z = 26)
c. p-electrons in Ne (Z = 10)
d. s-electrons in Mg (Z = 12)

Periodic Trends in Properties of Elements

5. From the following pairs of ions, which one is not an iso-electronic pair? *(2021)*

a. Na^+, Mg^{2+} b. Mn^{2+}, Fe^{3+}
c. Fe^{2+}, Mn^{2+} d. O^{2-}, F^-

6. For the second period elements, the correct increasing order of first ionisation enthalpy is: *(2019)*

a. Li < Be < B < C < N < C < F < Ne
b. Li < B < Be < C < O < N < F < Ne
c. Li < B < Be < C < N < O < F < Ne
d. Li < Be < B < C < O < N < F < Ne

7. In which of the following options, the order of arrangement does not agree with the variation of property indicated against it? *(2016 - I)*

a. Li < Na < K < Rb (increasing metallic radius)
b. $Al^{3+} < Mg^{2+} < Na^+ < F^-$ (increasing ionic size)
c. B < C < N < O (increasing first ionization enthalpy)
d. I < Br < Cl < F (increasing electron gain enthalpy)

8. The formation of the oxide ion, O^{2-} (g) from oxygen atom requires first an exothermic step and then an endothermic step as shown below:

$O(g) + e^- \rightarrow O^-(g)$; $\Delta_f H^o = -141$ kJ mol^{-1}

$O^-(g) + e^- \rightarrow O^{2-}(g)$; $\Delta_f H^o = +780$ kJ mol^{-1}

Thus, process of formation of O^{2-} in gas phase is unfavourable even though O^{2-} is isoelectronic with neon. It is due to the fact that, *(2015 Re)*

a. O^- ion has comparatively smaller size than oxygen atom
b. Oxygen is more electronegative
c. Addition of electron in oxygen results in larger size of the ion
d. Electron repulsion outweighs the stability gained by achieving noble gas configuration

9. The species Ar, K^+ and Ca^{2+} contain the same number of electrons. In which order do their radii increase? *(2015)*

a. $Ca^{2+} < Ar < K^+$ b. $Ca^{2+} < K^+ < Ar$
c. $K^+ < Ar < Ca^{2+}$ d. $Ar < K^+ < Ca^{2+}$

10. Be^{2+} is isoelectronic with which of the following ions? *(2014)*

a. Li^{+}
b. Na^{+}
c. Mg^{2+}
d. H^{+}

11. Which of the following orders of ionic radii is correctly represented? *(2014)*

a. $Na^{+} > F^{-} > O^{2-}$
b. $O^{2-} > F^{-} > Na^{+}$
c. $Al^{3+} > Mg^{2+} > N^{3-}$
d. $H^{-} > H^{+} > H$

Answer Key

1	2	3	4	5	6	7	8	9	10	11
b	c	c	a	c	b	c,d	d	b	a	b